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Chapter 7 Artificial Intelligence in Defense

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7.1 Introduction

Artificial intelligence (AI) has become a transformative force in defense, revolutionizing how militaries operate and respond to modern challenges. This chapter provides a comprehensive overview of AI's role within the defense sector, tracing its historical evolution from early experimental systems to the sophisticated technologies driving today's military advancements. Over the past 30 years, AI has progressed from niche applications to integral components of defense strategy, enhancing situational awareness, decision-making, and operational efficiency. By exploring key domains such as C2 systems, autonomous platforms, cybersecurity measures, and training environments, this chapter offers an in-depth

analysis of the significant contributions AI has made to defense capabilities and operations.

In addition to presenting the current state of AI in defense, this chapter also contextualizes the historical developments that have shaped its trajectory. From early research in machine learning (ML) and automation to the rise of complex human-machine interaction (HMI) technologies, AI's role in defense has expanded dramatically. This chapter outlines key AI-driven innovations and their implications for areas such as predictive systems, medical readiness, decision support, and cybersecurity, while also addressing the ethical challenges that accompany AI deployment. To provide a forward-looking perspective, the chapter proposes strategic frameworks to guide the responsible and effective use of AI, aiming to equip defense professionals and researchers with insights into both the achievements and future directions of AI in defense. Through this examination, the chapter offers a clear understanding of the objectives, scope, and transformative potential of AI within the defense sector.

7.2 Ethical Considerations and Challenges

AI in defense refers to the use of advanced computational systems that enhance decision-making, automate processes, and improve operational efficiency in military settings. AI encompasses a broad range of technologies, including ML, natural language processing (NLP), autonomous systems, and advanced analytics, which can process vast amounts of data to support situational awareness, optimize logistics, and enhance C2 capabilities. In defense, AI's scope extends across areas such as cybersecurity, uncrewed (autonomous) vehicles, intelligence analysis, training simulations, and many more applications. All are designed to provide a strategic advantage in complex and rapidly changing environments. This integration of AI into defense operations and training reflects its growing role as a critical asset in military strategy and effectiveness. It is critical that AI applications are both explainable (Arrieta et al. 2020) and trusted (Zhang et al. 2021).

The evolution of AI in defense has a rich history, beginning with early automation systems and progressing through decades of research in ML and

decision support. From basic rule-based systems in the twentieth century to today's sophisticated autonomous platforms and predictive models, AI technologies have continuously advanced to meet the changing demands of defense operations. However, alongside these advancements, the ethical implications of AI in defense have become increasingly prominent. AI's ability to make autonomous decisions, especially in life-or-death situations, poses unique ethical challenges that are unlike those faced in other sectors. The development of ethical frameworks and guidelines is essential to ensure that AI in defense is deployed responsibly, with considerations for accountability, transparency, and compliance with international humanitarian law. Addressing these ethical concerns is critical to fostering trust in AI-driven defense technologies and ensuring their long-term viability and reliability.

AI is transforming defense operations through applications in surveillance, autonomous weapons, cyber defense, logistics, and military training. By automating complex tasks, AI increases operational efficiency, accelerates decision-making, and enhances situational awareness through real-time data analysis and predictive capabilities. Autonomous systems reduce human risk by performing dangerous tasks, while AI-driven training simulations improve soldier preparedness and performance. These advancements offer significant benefits, including fewer human casualties and enhanced mission effectiveness, although they also raise ethical challenges that must be addressed for responsible deployment of reliable AI-based systems.

7.2.1 Ethical Principles in Defense AI

The integration of AI into defense raises significant ethical questions, particularly in the context of Just War Theory (Elshtain 1992), which provides a framework for evaluating the morality of warfare. Just War Theory's principles, such as *jus ad bellum* (the right to go to war) and *jus in bello* (the right conduct in war), must be applied to AI systems to ensure their use aligns with international ethical standards. AI-driven systems, especially autonomous weapons, must adhere to these principles by only being deployed in just causes and by ensuring that their actions in conflict respect proportionality and discrimination. Proportionality refers

to the need for military actions to avoid excessive harm relative to the military objective, while discrimination ensures that AI systems can distinguish between combatants and noncombatants. These principles are vital in preventing AI from being used in ways that violate ethical norms of warfare.

Accountability and responsibility in AI-driven defense operations present unique challenges. With AI systems making decisions, particularly in high-stakes situations such as targeting or mission execution, assigning accountability becomes complex. It is crucial to ensure that human operators remain in control and that AI systems are used as tools under human oversight. Commanders and decision-makers must retain responsibility for the outcomes of AI-driven decisions, and mechanisms must be established to ensure clear lines of accountability when errors or unintended actions occur. Ensuring that AI systems have clear constraints and that humans are involved in critical decisions, helps prevent a dangerous abdication of responsibility to machines.

Transparency and explainability are also essential ethical principles for AI in defense, particularly given the high stakes of military operations. AI algorithms must be transparent, allowing military leaders and policymakers to understand how decisions are made and to verify that they align with legal and ethical standards. Explainability is especially critical in life-and-death scenarios, where AI decisions about targeting or engagement must be fully understood by human operators to prevent mistakes and ensure moral accountability. Without clear, explainable decision-making processes, there is a risk that AI systems could operate in ways that are unpredictable or untraceable, eroding trust in their use.

Finally, minimizing harm is a key ethical priority in the use of AI in defense. AI systems must be designed and deployed in ways that reduce collateral damage and unintended consequences. This involves carefully balancing military necessity with ethical considerations to ensure that the use of AI does not lead to excessive harm to civilians or infrastructure. Strategies for harm minimization include building safeguards into AI systems, ensuring that autonomous weapons operate within strict ethical parameters, and continuously monitoring AI performance in the field to

correct any unintended consequences. By prioritizing harm minimization, defense organizations can ensure that AI is used responsibly and in alignment with both ethical norms and military objectives.

7.2.2 Challenges in Ethical AI Deployment

The use of AI in defense introduces a range of ethical challenges, one of the most prominent being the issue of bias and discrimination. AI systems, especially those used in military decision-making, are prone to inheriting biases from the data they are trained on or from the design choices made by developers. In defense operations, biased AI could perpetuate or even exacerbate discrimination, leading to unjust targeting or exclusion based on race, nationality, or other demographic factors. The risks of bias are particularly concerning when AI is used for surveillance, intelligence analysis, or autonomous targeting. To mitigate these risks, defense organizations must invest in the careful design, testing, and auditing of AI systems, ensuring that they are trained on diverse datasets and scrutinized for potential biases. Establishing ethical standards for AI development and continuously monitoring AI systems in the field is crucial for reducing the chances of biased outcomes.

The ethical debate surrounding autonomous lethal weapons is another critical challenge in AI's use in defense. Fully autonomous weapons systems, capable of selecting and engaging targets without human intervention, raise deep concerns about accountability and morality. One of the primary concerns is the potential loss of human control over life-and-death decisions, which could lead to actions that violate international humanitarian law. The absence of human judgment in critical moments may result in indiscriminate killing or excessive collateral damage. International laws and regulations have yet to fully address these issues, although efforts to establish guidelines for autonomous weapons systems are ongoing. The ethical dilemma centers on whether it is justifiable to delegate the decision to kill to a machine and how to ensure that autonomous weapons comply with the principles of proportionality and discrimination in warfare.

Cybersecurity concerns also present significant ethical implications in the context of AI-driven defense systems. AI has become a powerful tool in cyber warfare, capable of launching sophisticated attacks and defending critical infrastructure. However, the deployment of AI in cyber operations raises ethical questions about the protection of civilian infrastructure, privacy, and the potential for unintended consequences. AI-powered cyberattacks could disrupt essential services such as healthcare, energy, or financial systems, disproportionately affecting civilians. Moreover, the anonymity of cyber warfare makes it difficult to assign accountability for actions that violate ethical or legal standards. To address these concerns, ethical guidelines and international agreements must be developed to regulate the use of AI in cyber warfare and to ensure that military cyber operations prioritize the protection of civilian infrastructure and privacy.

The dual-use dilemma is another significant ethical challenge in AI for defense. Dual-use technologies refer to innovations that have both civilian and military applications. While AI research and development can drive innovation and enhance national security, there is a risk that these technologies could be misused by adversaries or non-state actors. For example, AI tools developed for civilian purposes, such as facial recognition or data analysis, could be weaponized in military contexts. Balancing innovation with security and ethical responsibility is essential to managing this dilemma. Governments and defense organizations must ensure that dual-use AI technologies are developed with ethical safeguards and that clear regulations are in place to govern their use, to preventing misuse while still fostering technological advancement.

7.2.3 Regulatory and Policy Frameworks

The development of AI in defense is governed by a growing body of international regulations and agreements, aimed at ensuring the ethical and responsible deployment of AI technologies. While there are currently no comprehensive international treaties specifically regulating AI in defense, several frameworks and guidelines influence its use. The United Nations has played a key role in shaping global discussions, particularly through initiatives like the Group of Governmental Experts on lethal autonomous weapon systems, which seeks to address the ethical and legal

implications of autonomous weapons. NATO also contributes to the governance of AI in defense through its policy frameworks and defense innovation initiatives, promoting the responsible use of AI across member states. Despite these efforts, international consensus on the use of AI in warfare remains elusive, with ongoing debates about the legality and morality of fully autonomous systems. The absence of binding agreements highlights the need for stronger international cooperation to regulate AI-driven military technologies and prevent misuse.

At the national level, several countries have developed policies and guidelines to ensure the ethical use of AI in defense. Leading nations such as the United States, the United Kingdom, and China have introduced national AI strategies that emphasize the importance of ethics and human oversight in defense applications. The US Department of Defense, for instance, has established AI principles that focus on responsible, equitable, traceable, and reliable AI development. These guidelines are designed to ensure that AI systems are used in ways that comply with both ethical standards and international law. The United Kingdom has also implemented policies that prioritize transparency and accountability in AI-driven defense operations, with a focus on minimizing unintended consequences (Montasari 2023). Case studies from these countries demonstrate that national strategies are crucial in addressing the specific ethical, operational, and legal challenges of AI in defense. However, differences in national policies can lead to fragmented approaches to AI governance, underscoring the need for harmonization across borders.

Industry standards also play a vital role in shaping the ethical deployment of AI in defense. Many private sector companies and industry organizations are leading initiatives to establish ethical guidelines for AI development, often working in collaboration with governments and academic institutions. These industry-led standards influence the design, testing, and deployment of AI technologies, ensuring that they meet ethical and safety benchmarks. For example, tech companies involved in defense contracts have increasingly adopted ethical AI frameworks that prioritize transparency, accountability, and human oversight. Moreover, collaboration between governments, industry, and academia has been essential in driving innovation while also embedding ethical considerations into AI

systems. By aligning industry standards with national policies and international regulations, stakeholders can create a more cohesive and responsible approach to AI development in defense, helping to prevent unintended harm and ensuring that AI technologies are used in compliance with global ethical norms.

7.3 AI-driven Innovations in C2 Systems

In this section, we describe AI and ML tools and methods that support military C2 and information, surveillance and reconnaissance (ISR) processes. The goal of C2 is to effect US policy in a complex, uncertain battlespace to “manage the tempo of battle, disrupt enemy actions, and assess a complex situation and strike the critical (not necessarily the most) at the right time” (Johnson and Levis 1989). C2, as a process, functions to support a military commander in (i) defining intent (e.g. goals and objectives), (ii) determining roles and responsibilities, (iii) establishing and complying with rules and constraints (e.g. schedules, rules of engagement [ROE]), and (iv) monitoring and assessing the situation as it progresses. (Alberts and Hayes 2006). ISR extends C2 to produce data products that inform C2 decision-makers. The ISR process, often referred to as TCPED, includes steps for sensor tasking, data collection, processing, exploitation, and data product dissemination (Tangredi and Galdorisi 2021). AI algorithms can support across the full range of these process steps.

C2 processes are complex and uncertain. From a computational perspective, C2 is a spatially distributed, parallel execution, multiobjective, open-world problem. C2 is distributed, in that plans are created and executed in multiple locations and at different hierarchical echelons, with varying levels of authority, specificity, and completeness. These plans are developed for and executed in the real world, which evolves over time and needs to be understood contextually from the perspective of a wide variety of actors. C2 process inputs range from dense (weather and terrain data) to sparse (local passive sensor detections) and from formally schematized (tracks in a common operational picture) to unstructured (fields in a hand-authored SPOTREP report or commander’s intent guidance document). C2 equipment and personnel, despite best efforts, rarely function in optimal conditions, and C2 data collection and plan execution

happen in areas where communications are denied, degraded, intermittent, and limited, all of which increases uncertainty and degrades mission synchronization and performance. C2 computational processes execute on a wide variety of hardware, including supercomputers, cloud compute platforms, shipboard server racks, desktop machines, small size, weight, and power (SWAP) compute nodes, and a wide variety of handheld devices, all of which, due to cost, have long refresh times such that they are often significantly behind the state of the art. Finally, C2 data, from an ML training perspective, is often difficult to obtain (due to collection cost and adversary denial activities), and C2 algorithms are often difficult to trust by human operators (due to black box behavior and functional inaccuracies, whether perceived or real).

Despite these challenges, and in some cases because of them, there is a great demand for improved AI/ML algorithms, models, and corresponding tools and decision aids. Over the last decade, significant investments in DoD and commercial research and development have produced new technologies that bring AI/ML-enabled C2 closer to common military deployment.

7.3.1 Planning and COA Generation

According to the DoD, C2 planning is an "orderly, analytical process, which consists of a set of logical steps to examine a mission; develop, analyze, and compare alternative courses of action (COAs); select the best COA; and produce a plan or order" (DoD 2020). From an AI perspective, planning is largely considered a search process within an n-dimensional space of plan steps, including environmental and adversary counter options and plan resources. A COA is a specific set of plan steps and resources, with evaluation criteria. AI algorithms can accelerate the planning and COA development process by analyzing available battlespace information (resource readiness assessment, threat assessment, weather impact analysis); optimizing the mapping of available resources to the COA evaluation criteria (target weapon pairing, sensor planning); accelerating the generation and evaluation of COAs against evaluation criteria (route planning, engagement planning); improving the identification of risks (ROE analysis, communications and logistics dependency analysis);

and automating the generation of work products (update briefings, C2 messages).

Over the last decade, improvements in these areas have come from increased compute resources (cloud computing centers, GPUs), the development of improved satellite-based communications and navigation methods, and the collection and digitization of increasingly large data sets, such as terrain and bathymetric data, satellite data collections, and open-source intelligence (warnings and indicators from real-time social media feeds, commercial imagery feeds, and financial transaction data).

AI algorithm improvements, leveraging these advances, have occurred in multiple areas. Three of particular interest are advances in distributed scheduling and task allocation, the application of reinforcement learning (RL) to planning, and the (as yet unproven) opportunity for leveraging large language models (LLMs) in the planning process.

7.3.1.1 Distributed Scheduling and Task Allocation

Distributed planning decomposes the COA search and evaluation problem across multiple participating nodes, such as members of an unmanned aerial system (UAS) swarm, members of a satellite constellation. In these tactical planning examples, local nodes have the best situation awareness and can use a variety of algorithms (e.g. market-based methods, stigmergy) to share information (including propagating uncertainty measures) and contribute to emergent global plans that are both efficient and resilient to degradation through communications or asset loss (Sauter and Bixler 2021). There is promise for such approaches to also support tactical planning for mosaic warfare, a new warfighting construct where a force package, such as a distributed kill chain including sensors, communications relays, and strikers, is composed on demand from available assets, instead of being planned in advance, based on estimates of availability and readiness (O'Donoughue et al. 2021).

7.3.1.2 Reinforcement Learning

In planning, like other AI application areas, the question of how the AI model is developed is key. RL has demonstrated that using models devel-

oped from participation in simulated game plans or observation of human gameplay can match or exceed human performance in a wide range of tactical and strategic planning tasks, such as fighter aircraft dogfight planning (Kawatsu et al. 2020) and counter UAS planning (Kawatsu et al. 2017b). The application of RL is challenged by the need to develop training simulations at a level of realism to lead to correct planning behavior, which is challenging to do due to the complexities mentioned above, and to provide adequate computational resources for the training process. In addition, the black box nature of RL models and the potentially novel or surprising decisions that come from simulation-based training can negatively impact operator trust in RL-based systems (Bunch et. al. 2024).

7.3.1.3 Large Language Models

The emergence of LLMs and associated algorithms and software architectures has generated considerable excitement, and anxiety, in the DoD planning world (Jensen and Tadross 2023). There are multiple potential applications of LLMs with the planning process, including generation of summary and analytic products that support diplomatic, informational, military, and economic analysis; automatic generation of wargaming models for COA analysis (Lebanoff et al. 2023); translation of computer format outputs of other planning analytics into human-readable text and graphic formats; and, when combined with other reasoning systems, actual COA generation or evaluation. As with other LLM applications in the military, there are considerable challenges to overcome, including the limits in the state of the art relative to complex, graphic-oriented, document to data ingest (maps, technical figures), the concern with inaccurate/hallucinatory outputs, the cost and SWAP of processing hardware, the challenges with applying LLMs within classified workspaces, the likelihood of bad actors intentionally executing supply chain attacks to poison LLM model development, and the potential for introducing new biases into the C2 analysis process.

7.3.2 Plan Monitoring and Risk Assessment

AI algorithms also have significant application in the emerging field of plan execution monitoring and risk assessment. Traditionally, plan execu-

tion monitoring and plan risk assessment processes were largely manual due to the lack of computational representations for plans and the complexity of manually encoding plans into a computational form. Both issues are solved by the advent of new AI planning tools, as noted above. Risk assessment occurs during the planning process, applying a mission, platform, or planning cell-specific risk policy to a generated plan and to the current and projected values of planning factors including weather, threat, and resource requirements. Execution monitoring extends this risk assessment process out of the planning process into the execution process to detect changes to mission risk that might require changes to the mission plan. Risk assessment can leverage the same AI methods as other parts of planning, but with emphasis on geospatial, temporal, and causal reasoning and risk profile knowledge representation methods.

7.3.3 Object Detection and Classification

Over the past decade, the performance of object detectors and classifiers has improved due to advances in deep learning (DL). Object detectors find instances of a certain class of object (human, airplane, and building) in the outputs of one or more sensors, such as electro-optical/infrared cameras or synthetic aperture radar. While further improvements in DL techniques are, synthetic data generation techniques are also advancing quickly to address DL's voracious appetite for training data. Synthetic data generation performs automatic training dataset generation using combinations of approaches, including applying physics-based rendering and realistic backgrounds to CAD models, using high-fidelity rendering engines, or using generative adversarial networks models.

7.4 AI Applications in Uncrewed Systems

The rapid advancement of AI has significantly transformed the uncrewed systems (UxS) landscape. AI is enabling greater levels of autonomy in air, ground, surface, and underwater systems, greatly increasing their utility and usability. Previously uncrewed air or ground robots had to be teleoperated or provided with simple plans to follow. With modern AI algorithms, sufficient intelligence can be embedded on the platform to remove the burden from the human. We will explore some of the ap-

plications for these autonomous systems, how AI is enabling this revolution, and what are some of the challenges that lie ahead.

7.4.1 Military Applications and Benefits

The lack of a human occupant means UxS can be smaller than manned platforms in size, weight, power, and cost to the point that hundreds or thousands of these systems could be deployed in an operation. These low-cost, expendable platforms can be placed in high-threat, denied environments where attrition is expected without putting humans in harm's way. They can adapt at computer speed to the adversary's actions, reacting faster than they can. But achieving this capability requires AI onboard these systems to accomplish their mission with minimal oversight by a human operator. Some of the missions envisioned for autonomous UxS include the following:

1. Surveillance and Monitoring: Autonomous UxS can cover vast areas over longer time periods more quickly and less expensively than manned systems. They can peer deep into urban canyons or around obstacles that can obscure traditional sensors. And they can be deployed closer to the fight, reducing the risk on warfighters.
2. Sensor Coordination and Fusion: Each sensor modality has unique positioning and coordination requirements to optimize their performance. Advances in swarm intelligence have demonstrated success in coordinating large multimodal sensor networks, combining imaging, radio frequency, and radar sensors into a single operational picture.
3. Persistent Target Tracking: UxS can form vast sensor networks to coordinate the tracking of multiple targets over large areas, cueing strike assets while maintaining target custody through the kill chain.
4. Coordinated Strike: when authorized by the human, autonomous UxS are able to perform coordinated strikes from multiple directions simultaneously, confusing and overwhelming enemy defenses.
5. Logistics Support: The movement of material and combat units consumes considerable time and manpower but is critical to the success of any military campaign. Commercial distribution concepts are being evaluated for military operations. AI and autonomous UxS can help

plan, coordinate, and move people and materials from distributed bases and weapons caches to the point of the spear (theater of war).

7.4.2 AI Technologies in Support of Autonomy

These and other missions require sufficient onboard intelligence to avoid dedicating operators to each platform. We briefly discuss a few of the key areas of onboard intelligence that enable this level of autonomy:

1. Perception: To be effective, the UxS must understand its environment, distinguishing friend, foe, and neutral. Recent advancements in image processing using deep neural networks (DNNs) enable UxS to analyze sensor feeds to identify objects or persons of interest. While training these detectors to be robust to obscurations and variations in appearance requires a lot of data, which is not always available, there have been promising advances in the use of synthetic data to train detectors to reach the proficiency required.
2. Planning and Decision-making: While the UxS is always operating under the guidance and direction of a human planner, it still needs to make onboard decisions to adapt to adversarial actions. Cognitive AI can reason like a human, determining how to get to a desired end state with the available resources and efficiently allocate tasking across the team.
3. Maneuvering and Control: Ultimately the UxS needs to decide where to go next and what to do in collaboration and coordination with other members of the team. DNNs have proven successful at controlling tactical aircraft in high-speed maneuvers of dogfighting (ref ADF article). Swarm intelligence has demonstrated success in coordinating the maneuvers of hundreds or thousands of UxS in military operations.
4. Human Interaction and Teaming: UxS will always operate under the supervision and control of the human. That requires the ability to process and understand the human planner's objectives and constraints and to communicate to the human what it is planning and why or to justify its actions. ML and swarm logic algorithms are opaque to human understanding, but cognitive systems excel in that department: both in processing and implementing the commander's intent and in explaining their actions using reasoning logic that hu-

mans can understand. This transparency greatly enhances trust in the system over time, which is essential if autonomous UxS are to operate as teammates with humans on the battlefield.

7.4.3 Challenges and Future Directions

While AI applications in unmanned systems offer numerous benefits, they also present challenges and opportunities for future development. We briefly discuss several areas that are critical to future system deployment.

1. Engineering, Testing, and Validation: Engineering the AI software that can control and coordinate the operation of dozens or hundreds of autonomous UxS is extremely challenging. New tools are needed that can support the development and testing of systems before deploying them into battle. Simulation tools, which can model large numbers of distributed systems and optimization and AI algorithms that can automatically test, analyze, tune, and validate autonomy software are essential.
2. Standards and Modular Open Systems Architectures (MOSA): The range of AI technologies and applications required to field autonomous UxS requires a whole-of-industry approach enabled by open standards and architectures. This allows small and large firms to contribute their unique technology toward a complete system without resulting in single vendor lock.
3. Ethical, Legal, and Social Implications (ELSI): The advent of autonomous UxS raises serious ELSI issues that need to be addressed. Can we engineer these systems to correctly distinguish friend, foe, and neutral and obey the relevant ROE and other constraints? Current rules on the use of lethal force need to be revisited when it's no longer a human who is threatened. These and other issues surrounding the use of UxS in wartime and peacetime operations need to be addressed.
4. Counter UxS Swarms: Our adversaries also recognize the importance of autonomous systems and are rapidly developing and deploying them without the same ELSI concerns or cumbersome acquisition systems. We need to develop inexpensive counter systems that have favorable cost exchange ratios to win the future fight.

7.4.4 Conclusion

The future of AI in autonomous UxS holds great promise for addressing the increasing challenges facing our military today. With AI, UxS can become true teammates with the warfighter, multiplying their effectiveness while reducing their risk. At the same time, our adversaries are catching up to us, and we need to rapidly increase our pace of development and deployment to maintain the slim lead we have in this race while not sacrificing our commitments to performance, safety, and ELSI constraints. The United States is still the world leader in AI, and we have the skills needed to win this fight, and it is one we cannot lose.

7.5 Application of AI to Cyber Operations

Current workflows for planning cyberspace operations are primarily manual and require significant resources. The pace at which adversaries innovate and launch attacks has been accelerated by improvements in AI, which makes it difficult to keep up. The digital nature of cyber warfare lends itself to automation, from lower-level digital artifacts to higher-level operational decisions. For example, more operational strategies can be explored by a single person through simulation against an autonomous adversary. An autonomous adversary is also useful in developing security operations or penetration testing to improve the efficiency of personnel. Cyberspace operators can use autonomous systems to augment their existing workflows and provide better coverage against the range of potential threats.

7.5.1 Representation of Cyber Tactics, Techniques, and Procedures

One of the requirements for tactical cyber systems is an understanding of the cyberspace domain. One of the frontrunners in this space is the MITRE ATT&CK framework (CITE). This framework characterizes a variety of cyberspace actions along with their prerequisites and outcomes and is a common representation throughout the industry to understand cyber operations. We have extended this framework by adding several layers of understanding and decision knowledge to form cyber tactics,

techniques, and procedures and defined their relationships to low-level artifacts and high-level objectives (Figure 7.1).

Figure 7.1 A stylized mapping of cyber tactics to individual artifacts that enable implementation of the tactics.

7.5.2 Cyber Attack AI Models

Models of the cognitive decision-making that drives cyberattacks incorporate the methods and reasoning of human attackers, including how they use various attack toolkits. The types of activities that human attackers use include, but are not limited to:

- Creating a map of the target system.
- Identifying potential vulnerabilities in the target system.
- Identifying the tools and attack approaches that can exploit the identified vulnerabilities.
- Creating specific attack goals.
- Choosing attack tactics relevant to the goals.
- Dynamically incorporating new information into the attack strategies, including encounters of newly discovered system information and observations of defensive measures or detection.

A key aspect of cyber attacker models is that they must be dynamic, adaptive, and innovative. There are often numerous possible approaches to attacking a system, each involving its own constraints and tradeoffs. In addition, the “best” choice often changes as new discoveries are made about the target system or as defensive countermeasures are employed. Cognitive models of cyber attackers incorporate these capabilities by implementing a rapidly executing decision loop that involves interpreting new information, understanding situational changes, selecting and revising goals, and adapting tactics to new goals and understanding.

Additionally, a hallmark of effective cyber attackers is their ability to innovate new attacks when they discover new types of vulnerabilities and/or new types of defensive measures. Every new defensive measure often incorporates its own tradeoffs, which an attacker can then use to exploit. As an example, if a defender has a tendency to block traffic from

suspicious ports, an attacker can exploit this tendency by spoofing otherwise essential services to make them appear to be suspicious. A cognitive attacker model capable of this type of innovation must be able to learn about defender responses and tendencies, analyze the consequences of those responses, and incorporate those consequences into new attack tactics that assist in achieving the attacker's goals.

One of the ways that a mature cyber action framework can be used is to learn effective cyberspace COAs. Cyberspace COAs are used in both offensive and defensive planning. We have developed research to use empirical game theoretic analysis (EGTA) to train offensive and defensive cyber agents in a simulated cyber network environment. EGTA features both game theory and RL and consists of repeated rounds of play where each agent takes actions based on an augmented MITRE ATT&CK framework described above. Game theory techniques are used to evaluate the result of each round, and RL is used to improve agents between rounds.

7.5.3 Cyber Defense AI Models

One of the prevailing issues in automated cyber defense is a deluge of data, which can result in a myriad of false positives. One of SoarTech's recent technologies, EpEx, is well-suited to explore the source of potential threats and provide evidence to confirm or deny the presence of a threat. EpEx uses episodic reasoning to provide evidence for hypotheses. This paradigm can be used to improve cyber defense by posing both positive (the system is compromised) and negative (a power failure caused suspicious data) hypotheses to explain observed behavior. We have designed a system built on the foundation of EpEx that gathers observations from a network, determines the extent to which they support various posed hypotheses, and compares the relative likelihood of each hypothesis to either raise an alarm or guide additional data collection.

7.5.4 Cyber Social AI Models

The advent of LLMs such as ChatGPT, Gemini, and LLaMA has revolutionized AI's capabilities in text generation and understanding. Trained on vast amounts of human-written text, these models can generate content

nearly indistinguishable from that produced by humans and comprehend complex written instructions. While these features make LLMs powerful tools for assisting with various tasks, they also present significant risks as spreaders of misinformation, false narratives, and targeted messaging.

LLMs are increasingly being utilized as the foundation for social media bots that can conduct large-scale information campaigns. By programming LLM agents to disseminate specific messages, adopt particular stances, and engage with targeted users, these bots can mimic social media users to promote certain narratives or sentiments. Such LLM-driven campaigns aim to influence real social media users' opinions and actions, often pushing them toward a desired political or social stance.

We have studied the dynamics of these LLM-driven influence campaigns using our synthetic social media environment, Griot. Griot allows us to create groups of LLM agents that interact within a controlled social media space, simulating various influence campaign scenarios. Each group is assigned different stances, behaviors, sentiments, and topics, such as supporting or opposing a political candidate or movement. For example, one group of agents may generate content in favor of a specific position and engage positively with like-minded agents, while another group is designed to oppose that stance and reply critically to sow dissent. This controlled environment enables us to test and observe how different influence strategies might play out in real-world social media settings, providing valuable insights into the tactics and impacts of such campaigns.

As agent-led influence campaigns become more prevalent, defending against them has become a critical concern. Various programs, such as DARPA INCAS, MIPS, and IWTSD's Indirect Influence Competition, are actively developing methods to counter these threats. These initiatives use AI-based tools to track, map, and mitigate influence campaigns, employing techniques such as ML algorithms, NLP, and network analysis to detect patterns of coordinated inauthentic behavior, identify agent-generated content, and trace disinformation back to its sources. These tools enhance the speed and accuracy of response efforts, offering scalable solutions to counter malicious activities across social media.

However, effectively combating these threats also requires robust social media intelligence (SOCINT) capabilities. SOCINT focuses on analyzing a targeted group's demographics, size, organizational structure, narratives, and network reach. While significant progress has been made in developing tools to analyze raw social media data, there is still a need for advanced post-analytic tools that help intelligence analysts incorporate SOCINT insights into their decision-making and reporting processes. Unfortunately, many analysts still rely on basic tools such as Microsoft Word and PowerPoint, which lack the necessary features for collaborative assessments or tracking complex data.

To address this gap, SoarTech has developed the Snapshot+ Suite, an AI-powered SOCINT toolset designed to enhance the effectiveness of intelligence analysts. Snapshot+ supports analysts in documenting their processes, investigations, and the provenance of the data they collect. It enables collaborative linking of social media accounts, classifications, groups, screenshots, and insights, storing this information in a graph-based model that maintains the structure of these linked concepts. This approach facilitates advanced sensemaking and nodal analytics, allowing analysts to move quickly from raw data to actionable intelligence and providing a more effective defense against LLM-driven influence campaigns.

By integrating AI-based defense programs with tools like the Snapshot+ Suite, we can better equip analysts to detect, understand, and counter the growing threat of coordinated influence campaigns, ultimately helping to protect the integrity of online spaces and the individuals who use them.

7.6 AI-enabled Training and Simulation Systems

In this section, we describe AI tools and methods in the context of developing more effective and efficient training applications, processes, and services. During training preparation, execution, and review phases, AI can contribute by informing, adapting, automating, optimizing, classifying/predicting, and generating training solutions that make training more effective and training processes more efficient. From an ethical point of view, AI-enabled training solutions should provide reliable services

where learner data is protected, performance predictions are accurate, and interventions (e.g. content selection and feedback) are both relevant and optimize learning and transfer of learning from training environments to defense operations contexts.

7.6.1 Training Preparation Process

During the preparation phase, AI acts as a strategic partner, transforming how training programs are designed and tailored to meet specific needs. By analyzing historical data, performance metrics, and learner profiles, AI tools and methods can be used to identify gaps in knowledge and skills, enabling a more customized and targeted training approach. For example, in a military training setting, AI can be used to analyze the results of past training sessions and individual performance data to suggest new content that focuses on identified weak areas, ensuring that the training is relevant and effective. Additionally, AI tools like NLP can assist in generating and curating up-to-date training materials, summarizing complex information, and even creating simulated environments that mirror real-world scenarios. This level of customization and precision ensures that the training is not only aligned with organizational goals but also highly engaging for trainees. In the following segments, we examine and provide training preparation examples of AI's impact on curriculum development, skill gap analysis, and optimal scheduling of training events.

7.6.1.1 Curriculum Development

The "tyranny of numbers" problem in computer chip development – where hard-wiring components became unsustainable as complexity grew – parallels the content bottleneck in adaptive training, where handcrafting the vast amount of content needed to make adaptive training work limits the scalability. Fortunately, current advances in LLMs allow us to generate the massive amounts of content needed for adaptive training (Choi et al. 2024; Leiker et al. 2023; Madunić and Sovulj 2024), creating content, assessment questions, and other instructional material needed. Additionally, learner context (e.g. performance or demographic data) can be used to customize the generated output of an LLM, providing bespoke content for the student at the time of need.

7.6.1.2 Skills Gap Analysis

To create a learner context, data from various sources can be utilized. AI-driven assessments identify knowledge gaps by building a detailed cognitive model, integrating with prior performance records, demographic data, and learner preferences. This holistic context allows adaptive training platforms to pinpoint specific gaps and tailor learning paths accordingly. When necessary, the platform generates content aligned with the student's goals, interests, and knowledge, ensuring personalized training that evolves with their needs. This approach enhances learning outcomes and scalability, providing each student with relevant, adaptive instruction that supports continuous improvement.

7.6.1.3 Scheduling and Resource Optimization

During the training preparation phase, AI revolutionizes the scheduling process by leveraging data-driven insights to optimize the timing and structure of training events. Traditionally, scheduling training can be a complex task, requiring coordination of numerous variables such as employee availability, workload, and resource allocation. AI simplifies this process by analyzing past attendance records, employee calendars, and performance data to identify the most effective times to hold training sessions. By predicting the optimal moments for scheduling, AI ensures that sessions are held when employees are most likely to attend and be engaged, minimizing disruptions to their regular duties and enhancing the overall effectiveness of the training.

AI also significantly improves the efficiency of resource allocation and capacity planning. Training programs often require specific resources, such as trainers, training rooms, and specialized equipment. AI tools can predict the exact resources needed for each session based on the type of training, the number of participants, and historical data. This capability prevents overbooking or underutilization of resources, leading to cost savings and a more efficient use of facilities. Furthermore, AI's ability to personalize training schedules according to individual learner needs means that employees receive training at times that align with their personal learning styles and job requirements, enhancing the relevance and impact of the training.

Beyond scheduling and resource optimization, AI brings a level of adaptability and automation to the preparation phase that was previously unattainable. Using predictive analytics, AI can foresee potential scheduling conflicts, such as unexpected absences or peak workload periods, and automatically adjust training schedules to accommodate these changes. This proactive approach ensures that training events run smoothly without the interruptions that typically arise from unforeseen issues. Additionally, AI can be used to automate the coordination and communication process, sending reminders, updates, and follow-ups to all participants. This reduces the administrative burden on training coordinators and ensures that participants are well informed and prepared, further increasing the likelihood of timely attendance and engagement. By harnessing the power of AI, organizations can create a more streamlined, effective, and efficient training preparation process that maximizes learning outcomes and minimizes resource waste.

7.6.2 Training Execution Process

During the execution phase, AI technologies take center stage, providing real-time adaptive learning experiences that respond to the needs of each learner (Figure 7.2).

Figure 7.2 Areas (in aqua) where AI influences real-time training process.

Midjourney, Inc.

Virtual instructors and AI-driven chatbots offer instant feedback and guidance, ensuring that trainees have the support they need at their fingertips. Other synthetic entities act autonomously in presenting military trainees with realistic scenarios. As trainees interact with each other and the simulation, AI is used to continuously monitor and access trainee behaviors and decisions (observational markers) to provide accurate assessments of trainee performance. This trainee decision-action process is captured in a trainee model that evaluates the trainee's progress toward mission and learning objectives. Finally, an AI model adapts intervention decisions by adapting either the content, the pace of instruction, or feed-

back to support engaging scenarios where trainee development is paramount.

An example of real-time AI-driven simulation is a medical training scenario. AI could analyze a trainee's performance in a simulated surgery and provide immediate corrective feedback or additional resources to address specific weaknesses. Moreover, AI can monitor engagement levels and performance metrics, such as time spent on tasks and accuracy, to dynamically adjust the difficulty of training materials. This adaptability ensures that all learners are adequately challenged and supported, optimizing both engagement and knowledge retention. Gamification elements introduced by AI further boost motivation, making the training experience more enjoyable and effective. In the following segments, we examine and provide training execution examples of AI's impact on adaptive learning, simulation and gamification, and AI-based instructors and simulation entities.

7.6.2.1 Adaptive Learning

Adaptive learning tailors training and educational experiences to individual or team attributes, optimizing learning, efficiency, and skill transfer (Sottilare and DeFalco 2020). Effective adaptive training solutions should be modular, extensible, and scalable. GIFT and STATS are two examples of modular solutions. GIFT exemplifies this by enabling personalized remediation in military counterinsurgency training, adjusting support based on real-time learner data (Spain et al. 2019). Similarly, STATS integrates adaptive training into existing systems, as demonstrated in a three-dimensional virtual training environment for basic electricity and electronics. Here, STATS enhanced the software with student modeling, adaptive interventions, and mastery visualizations, showcasing its flexibility and impact (Colby et al. 2024).

7.6.2.2 Simulation and Gamification

AI and gamification play a major role in today's DoD training and simulation systems. Taking cues from the commercial world of video games, both simple and complex AI systems control in game events, manage course materials, and imbue virtual role players with the ability to inter-

act with trainees in an unscripted manner. This all enhances realism and provides a more robust training experience. The algorithms behind these agents can range from simple recommender and matchmaking systems to complex machine-learned behaviors or SME-authored behaviors in cognitive architecture-based systems. These uses of AI can readily be seen across every domain, from linguistics to flight training and maintenance.

Taylor and Sims developed a system that models physical and cognitive behaviors of non-player characters in a cross-cultural training simulation. The AI in that system ensured that "the characters have the right physical appearance and speak the right language" and also that the characters "capture the richness of the interactions, the expectations that people from that culture would have about those interactions, and the responses to those interactions when the expectations are met or broken" (Taylor and Sims 2009). In a more recent example, Wray et al. (2015) created a system for specifying instructional content in a cross-cultural training system that is presented through adaptive agent-controlled NPCs that are more flexible and can interact with the trainee given unexpected inputs while still meeting training goals specified by the instructor.

Gamification, while not strictly an AI-enabled concept, works exceptionally well to maintain engagement in computer-based training programs and increase learning, and many serious games and simulations leverage gamification principles (McSorley 2022; Noh 2020). The term "gamification" comes from the introduction of various rewards for completing specific tasks or meeting goals. One could consider this an extension of the psychological concept known as "conditioning" where desired behaviors are rewarded and undesired behaviors are optionally punished (Akpan 2020; Alsawaier 2018). In this case, the desired behavior is maintaining engagement with the training system. All modern social and learning platforms use this to some degree, using badges, points, and notifications (both public and private). A well-documented use case in the language learning space is Duolingo, which uses rewards for completing language modules and completing translations; however, some say the platform goes too far with extensive use of negative nagging for users who have not interacted in a few days (Lashbrook 2020).

7.6.2.3 Intelligent Virtual Instructors and Virtual Entities

Virtual instructors employ cognitive modeling (Lee and Park 2008), RL (Mulian et al. 2024), and transfer learning (Sottilare et al. 2023) to facilitate pedagogical interactivity, automated feedback, and adaptive learning experiences. Intelligent tutors have recently been instantiated in the domains of military training (Sottilare et al. 2023), driver's education (Rehm et al. 2024), and financial literacy (Prasongpongchai et al. 2024).

Virtual entities have long been capable of simulating realistic and dynamic interactions for training purposes (Heinze et al. 2002), often deployed as highly defined state machine models or behavior trees. Advanced methods of behavior cloning, such as those using LLMs, have been applied atop these structures to create more sophisticated agent models (Lebanoff et al. 2023). Recent examples of AI agents include oppositional forces in military tactics simulations (Lebanoff et al. 2023) and synthetic pilots in air traffic control environments (Zuluaga-Gomez et al. 2023).

NLP enhances virtual instructors by enabling human-like, conversational interactions, resulting in real-time, responsive learning and personalized feedback (Alfehaid and Hammami 2023). Modern conversational instructor agents, such as Jill Watson, illustrate how NLP can be used to manage large volumes of student queries efficiently, providing accurate, context-specific responses derived from course materials (Kakar et al. 2024).

7.6.3 Training Review Process

In the review phase, AI's capabilities for data analysis and automation come into full play, driving continuous improvement and reinforcing learning outcomes. AI systems can automate the assessment process, providing instant, detailed feedback that helps learners understand their strengths and areas for improvement. For example, after a cybersecurity training session, AI could evaluate a participant's responses to simulated attack scenarios, immediately pinpointing where they excelled and where additional focus is needed. Beyond individual assessments, AI also aggregates data across sessions to identify trends and insights that trainers can use to refine their programs. By analyzing this data, AI tools can suggest modifications to training content and methods, ensuring they re-

main relevant and effective. AI can also schedule follow-up quizzes and reinforcement activities, enhancing retention and ensuring that learning continues beyond the formal training period. This comprehensive approach ensures that training programs are not static but evolve continuously to meet the changing needs of both the organization and its workforce. In the following segments, we examine and provide training review examples of AI's impact on performance analysis, feedback development, and interventions for continuous improvement.

7.6.3.1 Performance Analysis

Performance analysis in training review benefits significantly from detecting and modeling not just performance metrics, but also other learner states such as effort, engagement, and affect. AI-driven state detectors can analyze real-time data streams to identify when a learner is stuck in a mastery loop due to inadequate understanding (Colby 2020). By identifying these states, the system can predict future performance and other outcomes more accurately, leading to timely and optimal interventions. ML models such as logistic regression, decision trees, and neural networks are often employed to predict these outcomes, improving the adaptability and effectiveness of training (Aguirre et al. 2016).

7.6.3.2 Feedback Development

During the review phase of training, AI plays a transformative role in enhancing the effectiveness and efficiency of feedback development in after-action reviews (AARs). Traditionally, compiling and analyzing data from training sessions can be a time-consuming process, often delaying the delivery of critical feedback. AI changes this dynamic by rapidly processing vast amounts of performance data collected during training exercises – such as participant engagement, response times, and decision-making effectiveness. By analyzing this data in real time, AI generates immediate, objective feedback that is specific to each learner's performance. This enables trainers to conduct AARs promptly, focusing on the most crucial areas for improvement. For example, in a complex simulation exercise, AI could pinpoint exactly where a team's coordination broke down, providing precise feedback that helps participants understand what went

wrong and how to correct it. The immediacy and specificity of AI-driven feedback ensure that the review process is both relevant and impactful.

AI also significantly improves the quality and personalization of the feedback provided during AARs. Unlike traditional methods that might rely on general observations, AI tools can identify deeper patterns and trends that may not be immediately apparent to human trainers. By analyzing data across multiple dimensions, AI can highlight subtle improvements or declines in specific skills, offering more nuanced insights into each participant's performance. Moreover, AI can tailor feedback to individual learners, taking into account their unique learning styles, past performance, and areas of growth. This personalized approach makes the feedback more meaningful and actionable for each participant, increasing the likelihood of positive behavior changes and skill development. Participants receive feedback that speaks directly to their experiences and needs, making the review process more engaging and effective.

AI can also be used to streamline the entire AAR process by automating the generation of comprehensive reports that summarize key findings and recommendations. These AI-generated reports can include visual aids such as charts and graphs that clearly illustrate performance trends and highlight areas needing attention. This automation not only reduces the time and effort required for trainers to manually compile and analyze data but also ensures that the feedback is presented in a clear, understandable format. Trainers can then focus more on facilitating productive discussions and developing actionable improvement plans based on AI-driven insights. Furthermore, AI can suggest specific follow-up actions or learning resources tailored to each learner's needs, ensuring that the feedback provided during AARs leads to meaningful, sustained improvements. By making the review process more efficient and insightful, AI helps organizations maximize the effectiveness of their training programs, driving continuous improvement and better outcomes.

7.6.3.3 Interventions for Continuous Improvement

During the review phase of training, AI can be instrumental in continuously improving interventions such as feedback and content adaptations

by leveraging its ability to learn from data and refine strategies in real time. As AI systems analyze performance metrics, learner engagement, and feedback effectiveness, they can identify patterns that indicate which interventions are most successful in promoting learning and retention. For instance, if AI detects that certain feedback formats or content adaptations consistently lead to better outcomes for specific learner profiles, it can automatically suggest or implement similar adjustments for future training sessions. This continuous loop of analysis and adaptation allows AI to fine-tune the training process, ensuring that interventions are increasingly targeted and effective. By dynamically adjusting the training content and feedback based on real-time data, AI helps create a more responsive and personalized learning environment that evolves to meet the changing needs of learners, ultimately enhancing the overall effectiveness of the training program.

AI can effectively improve the generation and selection of trainee interventions (e.g. content changes or feedback) during the training review phase. Several researchers have shared the transformative impact of advanced technologies in training programs. Hurd et al. (2019) underscored that iterative feedback and evaluation are critical for effectively improving the effectiveness and efficiency of training programs. In iterative feedback and evaluation research, Wilson et al. (2021) demonstrated how AI methodologies enhance benchmarking practices in training by offering precise evaluations and improvements (Wilson et al. 2021). Adding to the knowledge base, Green et al. (2020) depicted adaptive learning technologies that utilize real-time data to personalize training (Green et al. 2020). This continuous influx of new data enables developers to minimize drift and keep AI-based models used for intervention generation and selection optimized. Kumar et al. (2022) added to these findings, illustrating how ML approaches can further refine performance benchmarking in educational settings (Kumar et al. 2022).

7.7 AI-enabled HMI Technologies

HMI technologies are defined as tools, systems, and methods that enable communication between a human user and a machine. This includes a wide range of devices that allow a user to control, interact with, and re-

ceive feedback from computers and other digital systems. Broad advances have been made in the development of HMIs from early designs of mouse (Engelbart 2023) and keyboards. More recently, advances in AI technologies have contributed to enhancing and improving overall user experience, making interactions more intuitive and personalized, which in turn drives efficiency. In this section, we examine these impacts as they relate to the type and quality of interaction with the user and describe and give examples of how AI has been used to improve HMI technologies, specifically in a defense context.

7.7.1 Intuitive User Interfaces

Intuitive AI systems are designed to make interfaces more natural by allowing the user to interact with the technology in a similar fashion to which they interact with other people. This design approach applies across user input, the presentation of information to the user, and system adaptation or feedback based on tracking user and environmental state.

Examples of more intuitive user input for HMIs include the advancement of speech recognition along with NLP that allow systems to identify proper phraseology and speech patterns, allowing conversational AI, context-aware interactions, and real-time multilingual translation (Folsom-Kovarik et al. 2022; Hung et al. 2020; Tanaka et al. 2024). Both of these approaches support more natural, intuitive interaction between a user and system and allow for efficient command execution in both a training and operational defense setting.

One example is utterance recognition applied to Navy shipboard radio operator training, in which two-way radio communications are simulated and graded for naval students required to provide verbatim radio responses to a prompt (Ulinski et al. 2024). More generally, multimodal interactions that have benefitted from AI include gesture recognition, touch, visual, and face recognition. AI algorithms can be used to process these naturalistic interactions and fuse multiple modalities in real time into a single, coherent response.

Examples of intuitive presentation of information to the user within HMIs at the tactical level include helmet-mounted displays, ground con-

trol UAV stations, and AR/VR interfaces, among others. AI can enhance AR/VR experiences by recognizing objects in the environment, predicting user actions, and adapting the virtual space based on user inputs. This allows for immersive training simulations such as those in development for combat medics (Tanaka et al. 2019) and AR trainers in which intuitive overlays can be provided by an AI assistant (Taylor et al. 2024). LLMs have provided the capability to understand and generate human-like text and have been applied in both conversational interfaces and chatbot systems along with intuitive user interfaces that can process and summarize large amounts of data to the user in a condensed fashion on a tactical display.

One such tool in development as part of DoD Health CONOPs is a global health engagement user interface that is designed to plug into existing tactical battle management displays, e.g. Joint Operational Medicine Information Systems Medical Common Operating Picture (MedCOP; PEO Defense Healthcare Management System 2025), with an HMI that allows a user to obtain a LLM-based summary of tactical-level hospital data and patient distribution options in response to a crisis or conflict using speech or text-based query (Sussman-Fort et al. 2025). The presentation of information can also be optimized by DL approaches applied to cognitive state assessment. This has been used in pilot heads-up displays for triaging the presentation of critical flight information to pilots (Lasko et al. 2024).

More generally, AI has been used to predict and automatically determine cognitive workload (the utilization of working memory while performing a task) based on electroencephalography (EEG) and other peripheral physiologic measures (eye-blinks, pupil dilation, heart rate variability) to optimize the intuitive presentation of key information and minimize distraction within an HMI system.

7.7.2 Personalized HMIs

Personalized HMIs use AI to learn how people prefer to interact or have information delivered and then adapt or tailor responses to user preferences. The idea is that adapting the modality of user input and the presentation of information to a user can both speed up and optimize HMI

performance. Examples include AI applied to eye-tracking and gaze detection, multimodal data fusion, and emotion recognition. AI can process eye-tracking data to infer user intentions and adjust system behavior accordingly.

AI applied to multimodal data fusion can enable context-aware interactions, allowing HMIs to interpret user intent more accurately by combining speech and gesture, for example. Emotion-aware systems apply AI to analyze physiologic data, facial expressions, and body language to adjust system responses for a tailored, engaging interaction. DoD applications include (i) adaptive training in which AI systems track performance metrics, reaction time, accuracy, and decision-making patterns to adjust the presentation of information in terms difficulty level, pace, and content of the training in real time (Tanaka et al. 2024; Colby et al. 2024); (ii) autonomous vehicles such as the DARPA Sea Hunter unmanned surface vehicle and the Army's Robotic combat vehicle in which AI is used to tailor the HMI by adjusting vehicle control interfaces based on operator input patterns and mission needs including predicting operator intent to offer semi-autonomous or fully autonomous actions as needed; (iii) National Geospatial Agency geospatial intelligence systems, which can adapt to an analyst's search patterns, suggesting probable areas of interest or potential threats based on historical data and real-time eye-tracking or physiologic inputs (NGA 2024); and (iv) wearable health monitoring systems that analyze physiologic data to detect early signs of fatigue, dehydration, or injury and provide adaptive feedback to the soldier and command center while learning individual health patterns, enabling more personalized health recommendations and alerts during missions.

7.7.3 Future Directions for Intuitive and Personalized HMIs

Together, intuitive and personalized AI-driven HMIs drive efficient communication and interaction by providing natural user inputs and tailored presentation of information to the user while adjusting based on the user and environment state. Moving forward, it is likely that the future of AI-driven HMI systems will expand upon current adaptive approaches to continuously improve their interaction based on user feedback and evolving operational needs. As defense systems become integrated across

domains and services, there will be a need for standardized HMIs that can operate different platforms and devices and can automatically adjust based on situational awareness and context while presenting information to the user that is clear, transparent (Gruber et al. 2024; Barber et al. 2024), accurate, and understandable.

Current HMIs are limited in providing the operator situational awareness of system status and processes, status, and operational context (DARPA 2024). The result is that the user and system can be at odds with each other during operation, potentially leading to catastrophic failure such as that seen with the Boeing 737 Max Maneuvering Characteristics Augmentation System (FAA 2024). Recent work on the DARPA EDGE program sought to create HMI design tools and "build HMI systems that allow operators to not just monitor autonomous systems but also adapt their use to meet the needs of unanticipated situations" (DARPA 2024).

Future approaches include employing the theory of graceful extensibility into the HMI design process by applying explainable AI (XAI) to increase the resilience and adaptive capacity of an HMI system, or the ability to adapt to unexpected circumstances in the face of catastrophic failure and any perturbance that impacts operation (Woods 2018). HMI systems will also have the ability to be increasingly tailored to the individual warfighter by applying advances in the application of brain-computer interfaces (BCIs). AI and DL applied to brain measurements using less invasive measures such as EEG and electrocorticography have opened up the possibility for intent-based communication via imagined motor or speech actions (Garcia-Salinas et al. 2019; Watanabe et al. 2020). This would allow a system to be tailored or trained to a user's brain activity using DL or other ML approaches and be actively controlled by an imagined intent without any physical action. Balancing this type of active BCI are so-called passive BCIs that use AI algorithms to adapt to default-mode networks in the brain and enhance key user features based upon user state and background context. The result is an increase in the signal-to-noise ratio in HMIs with the corresponding ability to respond immediately and appropriately to changes in the situational environment using automated brain processes for deviance detection without top-down user input (Sussman et al. 2014). Together, increasingly intuitive and personalized AI-en-

hanced HMI systems will bring a unified approach to supporting warfighter-machine/system communication across the DoD.

7.8 Integrating Machine Reasoning and Explanation for Dynamic Decision-making

Intelligent systems are dynamic, adaptive, and self-reflective, capable of reasoning about their actions and adjusting to changing environments (Hayes-Roth 1995; Franklin and Graesser 1996), whereas conventional software systems are rigid, procedural, and operate without understanding or adapting to the reasons behind their predefined tasks (Doyle 1979). Through our decades of work developing intelligent decision-making systems, combined with lessons learned from other researchers, practitioners, and developers, we have identified an abstract framework for developing such intelligent systems.

Central to this framework is the integration of both reasoning-based architectures and ML components, creating hybrid systems that leverage the strengths of statistical learning models for perception and sensemaking while relying on cognitive architectures for dynamic goal setting, reasoning, and adaptation. This hybrid approach enables intelligent systems to not only generalize knowledge from data but also to apply that knowledge in a reflective and context-sensitive manner, addressing new tasks and adapting to unfamiliar situations in ways traditional systems cannot.

One of the fundamental keys to this framework is that intelligent decision-making depends crucially on the system understanding *why* it is doing *what* it is doing. This reflective capacity, often formalized through meta-reasoning architectures (Kotseruba and Tsotsos 2020), enables the system not only to execute actions but also to introspectively evaluate its own decision-making process. Furthermore, intelligent systems differ from traditional software in their ability to leverage existing knowledge to generalize and address new tasks. While the software developer understands why the software system must execute the instructions being defined, those reasons are not explicitly represented or reasoned about inside the software system itself. In contrast, intelligent systems can adapt

to new, unfamiliar situations by acquiring new knowledge, representing it flexibly, and generalizing it to different contexts (Laird 2012).

Figure 7.3 illustrates an abstract model of intelligent decision-making, structured around a continuous feedback loop with the environment. This model, which is a practical application of the belief-desire-intention (Bratman 1987; Georgeff et al. 1999) paradigm, reflects one example process through which intelligent systems can perceive their environment, update their internal states, and execute actions.

Figure 7.3 Abstract model of intelligent decision-making in environments.

This cycle begins with *sensing the environment*, where the system gathers new information from external sources, allowing it to maintain situational awareness. Following this, the system *revises its beliefs*, updating its internal representation of the world (e.g. working memory; see Jones and Wray 2006) based on the sensed information. These beliefs represent the system's current understanding of its situation, forming the foundation for its reasoning processes.

Next, the system *Revises its desires*, which involves adjusting what it aims to achieve based on the newly revised beliefs. Desires represent the system's *aspirational states* – outcomes it would prefer to bring about. Next, the system *revises active goals*, selecting which desires it will actively pursue based on their feasibility and priority within the current environment.

Once the active goals are updated, the system *selects or reconsiders its plan* – choosing or adjusting a course of action sequence that best aligns with the set of active goals, balancing available resources and environmental conditions. The system then *executes the planned action*, performing the chosen actions to interact with the environment to achieve (or attempt to achieve) the goals it has set. This process is dynamic, constantly adjusting its course as new information is sensed and new decisions are made.

In contrast, traditional software systems interact with their environment in a far more rigid and predefined manner (Newell 1992). These systems typically follow a fixed sequence of input processing, without the ability to dynamically reason about or adjust their behavior in response to changing conditions. When new information enters the system, it processes the data according to a predetermined algorithm or set of rules, often without the capacity to evaluate the broader context of that information or update its understanding of the environment. Once inputs are processed, conventional software executes a predefined set of actions based purely on hardcoded logic or scripts, unable to reconsider goals, revise plans, or adapt its behavior if the situation changes.

Modern software often integrates ML or statistical learning models to enhance performance in specific tasks such as perception, classification, or prediction. While ML systems can generalize across new data points within the scope of their training, they do not possess an explicit representation of beliefs, desires, or goals, which are central to the reasoning process in intelligent systems. Instead, ML systems rely on statistical relationships encoded in the model, focusing on pattern recognition and inference without an understanding of why certain decisions are made. This lack of explicit reasoning results in systems that cannot introspect or explain their decision-making process in an understandable way. When explanations are needed, they are often post hoc; that is, plausible rationalizations are constructed after the decision is made, rather than arising from a structured decision-making process (Rudin 2019).

Additionally, while ML systems can perform well on narrowly defined problems, they often struggle to generalize to entirely new tasks or domains without extensive retraining. This is because their learning is typically constrained by the specific data on which they were trained, limiting their adaptability beyond those contexts. In contrast, cognitive architectures like the one described earlier incorporate a reflective capacity that allows for dynamic goal revision and flexible reasoning across tasks.

To address these limitations, our system design often implements *hybrid architectures* that integrate ML algorithms within a broader intelligent decision-making framework. In this layered approach, ML models, such

as convolutional neural networks (CNNs), are used for perception and environmental sense-making, while traditional intelligent reasoning structures handle goal-driven reasoning, decision-making, and adaptation (Sheth and Thirunarayan 2021). For example, we have applied CNNs for gesture recognition within complex cognitive systems (Kawatsu et al. 2017a, 2017b), Q-learning algorithms for goal selection and adaptive decision-making (Newton et al. 2022), and ML-based behavior cloning techniques to assist in plan selection (Lebanoff et al. 2023). These hybrid systems, which blend the strengths of statistical learning and cognitive reasoning, are often referred to in the literature as third-wave AI (DARPA 2017), designed to address both the need for adaptive reasoning and the computational efficiency of ML.

By leveraging both paradigms, these systems are capable of handling complex environments that require not only data-driven pattern recognition but also the capacity to reason about the world, explain decisions, and adjust goals and strategies in response to changes. This hybrid approach is particularly well-suited to dynamic and open-ended domains where adaptability and introspective reasoning are essential for success.

Intelligent systems represent a significant advancement over traditional software and ML approaches by incorporating dynamic reasoning, goal revision, and the ability to generalize across tasks. These systems, through their reflective architectures, can adapt to changing environments, introspectively assess their actions, and explain their decisions in meaningful ways. By integrating ML techniques for perception and recognition with intelligent reasoning frameworks, we create hybrid systems that can address complex, evolving challenges with both computational efficiency and adaptive decision-making. This synthesis of learning and reasoning provides a robust foundation for the next generation of intelligent systems, equipped to thrive in dynamic and uncertain domains.

7.9 Responsible AI in Predictive Systems and Medical/Defense Health Readiness

AI is poised to play a transformative role in healthcare by enabling more accurate and timely diagnosis, personalized treatment plans, and opera-

tional efficiency. Through ML and data analytics, AI systems can analyze vast amounts of medical data, identifying patterns and trends that might be overlooked by humans and forecasting patient outcomes, allowing healthcare providers to take proactive measures to improve the quality of care while optimizing resource allocation. In the context of the military's healthcare needs, AI solutions can support battlefield medicine by enabling faster triage and diagnosis through real-time data analysis and predictive insights, ensuring warfighters receive the most appropriate care when and where they need it. Yet, as with any transformational technology, it is important to understand not only the *technical* challenges attendant with large-scale adoption of AI but also the *responsibility* challenges surrounding its adoption.

7.9.1 ELSI: A Framework for Responsible AI in Defense Healthcare

The *responsibility* challenge begins long before an AI tool is ready to be delivered to users, starting at the earliest stages of conceptualization, when developers are not locked into a single design approach. It is at this early stage that consideration of key ELSI associated with developing and deploying an AI tool is most crucial. Much like DevSecOps (Development, Security, and Operations), an approach that integrates security practices as a shared responsibility across the entire software development lifecycle (SDLC), the ELSI framework emphasizes the need for responsible, ethical, and transparent AI systems throughout the entire SDLC. Military healthcare AI systems must address concerns about bias, the reliability of predictive models, and the balance between human judgment and machine automation to maintain trust and operational effectiveness in resource-constrained and high-stakes environments. Properly implemented, ELSI considerations ensure that AI systems mitigate these concerns.

7.9.2 Context of Defense Healthcare Systems

The Department of Defense has embraced a warfighting strategy called "distributed operations," in which military assets are dispersed across wide areas, allowing teams to quickly adapt to local conditions without waiting for centralized direction, increasing their flexibility and ability to

respond to dynamic threats or opportunities. This approach complicates communication, coordination, and delivery of medical care, as medical units can be isolated, resources can be widely distributed, and evacuation can be challenging to plan and implement.

7.9.3 Applying AI in Defense Health Systems

AI tools can address these challenges across the entire *care continuum* – a patient flow process that encompasses the series of progressively formalized medical interventions. The continuum begins with care delivered at the point of injury by first responders (role of care 1), moving to care provided at mobile field hospitals (role of care 2), progressing to theater hospital care that provides intensive care and surgical services (role of care 3), and ending with definitive care provided at hospitals and treatment facilities well outside the theater of operations (role of care 4). At the earlier stages of the continuum, where time is critical, resources are limited, and decision-making occurs under constant stress, AI-enabled tools can be a force multiplier by providing just-in-time training on little-used procedures, triage assessments, and prioritized treatment recommendations. At the later stages of the continuum, these tools can streamline administrative tasks like managing patient records and optimizing hospital workflows.

7.9.4 ELSI Considerations with AI in Defense Health Systems

Regardless of the role of care for which they are developed, these systems share a common design approach that includes selecting the appropriate data sources to develop and train the algorithms underlying their AI and presenting in an understandable manner the outputs from those algorithms. We consider ELSI in terms of three stages in the design and fielding of healthcare tools: data collection, AI algorithm development, and user interactions.

Data are the foundation from which the algorithms driving AI solutions are developed, and ensuring data quality and integrity is paramount. Ethical considerations demand that these data accurately represent the populations these tools will serve so that the resultant tools will be broad-

ly applicable. Legal considerations require that the collected data be safeguarded against uses other than algorithm development. Social implications of low-quality data include tools that deliver inaccurate results, creating mistrust and slowing the adoption of AI tools.

Ultimately, humans design the algorithms driving AI tools, and human developers' assumptions, values, or preferences can be embedded in the algorithm's rules, leading to algorithm bias. Ethically, these biases can lead to unfair apportioning of resources or treatments – especially in the early stages of the care continuum. Legally, these biases can violate antidiscrimination laws, especially if the bias arises in a demonstrably systematic fashion. Socially, biased AI can worsen inequalities and reduce trust in defense health systems and the broader military mission.

Particularly in roles of care 1 and 2, military healthcare providers, under stress, time constraints, and the threat of injury, might over-rely on AI tools to quickly make critical decisions in high-pressure situations. Ethically, this can undermine the role of human expertise in the human-AI team, leading to poor outcomes that might otherwise have been avoided. Legally, this creates uncertainty about liability when AI makes incorrect predictions, blurring the lines of responsibility. Socially, over-reliance can diminish overall trust in the military healthcare enterprise and reduce morale and "the will to fight."

7.9.5 Mitigations

Implementing ELSI as a living process across the development cycle provides a manageable and traceable approach to mitigating these challenges. Data quality and integrity can be ensured by implementing robust data validation techniques, implementing regular audits of input data, and standardizing data formats and collection protocols. Algorithmic bias can be reduced by diversifying training datasets and monitoring AI outputs for disparities across different demographic groups. Finally, over-reliance on AI tools can be mitigated by including explainability features, allowing users to understand the basis for its predictions, and by training personnel to question AI outputs and use them in conjunction with their expertise.

7.9.6 Summary

AI has the potential to transform healthcare, particularly in military settings, by enabling faster diagnoses and operational efficiency. Realizing this potential requires that ELSI must be considered throughout the AI development process to avoid bias, protect data integrity, and maintain the balance between human judgment and AI recommendations. Considering the ELSI framework throughout the AI development cycle can mitigate these challenges.

7.10 Future Directions

The future of AI in defense applications is one of continual innovation and increased autonomy. As AI technologies mature, their integration into defense operations will not only enhance operational efficiency, situational awareness, and decision-making but also ensure that defense forces are better prepared for the challenges of tomorrow's battlefields.

7.10.1 Future Directions in C2 Systems

The future of AI-driven applications in defense C2 systems is poised for significant advancements, with innovations focusing on enhancing real-time decision-making and situational awareness. AI technologies will increasingly integrate vast amounts of sensor data, automating the analysis and synthesis of information to provide commanders with actionable insights in complex, multi-domain environments. Future C2 systems will likely incorporate advanced AI algorithms capable of predicting threats and automating routine tasks, enabling human operators to focus on strategic decisions. Innovations in AI will also enhance the interoperability of these systems, facilitating better coordination among allied forces and improving response times in fast-paced operational theaters.

7.10.2 Future Directions in UxS

In the realm of uncrewed/autonomous systems, AI will drive advancements in autonomous navigation, decision-making, and mission execution. Future innovations will enable these systems to operate with greater

independence and flexibility, allowing them to perform complex tasks with minimal human intervention. AI will enhance the autonomy of aerial, ground, and maritime platforms, enabling them to adapt to changing conditions and environments. These advancements will also focus on improving communication and coordination between autonomous systems and human operators, fostering more seamless and efficient teamwork in military operations.

Autonomous operations will continue to evolve, with AI enabling the automation of increasingly complex tasks across a variety of defense platforms. Future autonomous systems will be capable of executing multistage missions with minimal human input, from reconnaissance and surveillance to direct engagement with adversaries. These systems will operate in swarms, where multiple uncrewed vehicles work collaboratively to achieve mission objectives, improving operational efficiency and reducing risk to human forces.

7.10.3 Future Directions in Cyber Operations

AI-driven cybersecurity systems will continue to evolve, leveraging ML and DL techniques to detect, analyze, and counter cyber threats with unprecedented speed and precision. Future cybersecurity systems will incorporate predictive analytics, enabling defense organizations to anticipate potential cyberattacks and mitigate risks before they materialize. AI will also facilitate the development of autonomous cyber defense mechanisms that can respond to attacks in real time without human intervention. As cyber threats become more sophisticated, AI-driven cybersecurity systems will be essential in protecting critical defense infrastructure, military networks, and civilian assets.

Cybersecurity innovations will continue to focus on developing proactive and adaptive defense mechanisms. Future AI-driven systems will evolve to counter increasingly sophisticated cyber threats, with AI playing a critical role in developing autonomous response strategies. These systems will detect and neutralize emerging threats in real time, protecting both military and civilian infrastructure from disruption. Advances in AI will

also improve the ability to identify and mitigate insider threats, protecting sensitive defense information and assets from unauthorized access.

7.10.4 Future Directions in Training and Simulation

Training and simulation systems will benefit from AI's ability to create highly realistic, adaptive learning environments. Future AI-driven training systems will offer personalized and scalable training solutions that adjust in real-time to the needs and performance of individual soldiers. These systems will integrate virtual and augmented reality (AR) to provide immersive experiences, enhancing the transfer of skills from training to operational contexts. AI's ability to model complex scenarios will allow for more dynamic and unpredictable training environments, preparing soldiers for a wider range of potential situations and adversaries.

Specifically, adaptive learning is the most promising area for future AI innovation and will revolutionize military education and skill development. AI-powered systems will offer continuous, personalized learning experiences that adjust to the evolving needs of individual soldiers and defense personnel. These systems will track progress over time and adapt their content and training methods to ensure optimal learning outcomes. As a result, soldiers will benefit from more efficient and targeted training programs, enhancing their readiness for complex operational challenges.

7.10.5 Future Directions in HMI Technologies

Future directions in HMI for defense will focus on creating more seamless and intuitive interactions between humans and AI-driven systems, ensuring that soldiers and commanders can efficiently control and interpret complex technologies. One of the key innovations will be the development of adaptive interfaces that respond to the cognitive and physical states of the user. For example, AI-powered systems will be able to monitor a user's stress levels, fatigue, or focus and adjust the interface to simplify decision-making or provide additional support when needed. This adaptability will be crucial in high-pressure environments, such as combat operations, where quick and accurate decision-making is critical. Wearable technologies, such as AR glasses and haptic feedback devices,

will further enhance these interfaces, enabling users to interact with AI systems in more natural and immersive ways.

Another major area of advancement will be the integration of advanced NLP capabilities into HMI. Future AI systems will allow defense personnel to communicate with machines using voice commands and natural speech, streamlining interactions and reducing the need for specialized training to operate complex systems. These NLP-driven interfaces will enable more efficient communication between human operators and AI-driven systems, especially in C2 settings, where real-time decisions must be made quickly and accurately. Additionally, AI will play a role in interpreting nonverbal cues, such as gestures or eye movements, further enhancing the fluidity of HMI. These advancements will make AI-driven systems more accessible, intuitive, and user-friendly, significantly improving the collaboration between humans and machines in defense operations.

7.10.6 Future Directions in Machine Reasoning and Dynamic Decision-making

Future directions in machine reasoning and decision-making in defense will focus on creating AI systems that can handle increasingly complex and dynamic operational environments with greater autonomy. As AI technologies advance, machine reasoning will evolve from basic rule-based systems to more sophisticated models capable of understanding and interpreting intricate, multilayered scenarios. These systems will be able to synthesize large volumes of real-time data from various sources – such as satellite feeds, sensors, and human inputs – and use advanced reasoning techniques to generate optimal COAs. For instance, AI will be able to assess battlefield conditions, predict enemy movements, and recommend tactical strategies in near real-time, allowing military commanders to make more informed decisions under rapidly changing circumstances. This shift toward more autonomous and context-aware reasoning systems will greatly enhance decision-making speed and accuracy, reducing the cognitive load on human operators.

Another area of immense progress in machine reasoning will be the continued development of XAI systems, which address one of the critical

challenges in military AI applications: transparency and trust. Future AI decision-making systems will not only generate recommendations but also explain the rationale behind their conclusions in a way that is understandable to human operators. This capability will be particularly important in high-stakes scenarios where human oversight is necessary to validate and approve AI-driven decisions. By improving the transparency of machine reasoning processes, XAI will build greater trust between human operators and AI systems, fostering better collaboration and ensuring that autonomous decision-making remains aligned with ethical and strategic goals. Additionally, the integration of machine reasoning with predictive analytics will further improve the AI's ability to anticipate future events, creating a more proactive decision-making framework that enhances military preparedness and response.

7.10.7 Future Directions in Defense Healthcare

In medical and defense health readiness, AI will play an increasingly prominent role in predictive health monitoring, diagnostics, and treatment. Future AI-driven systems will offer real-time health assessments for soldiers, predicting potential health issues and recommending preventive measures to ensure optimal physical and mental readiness. AI will also assist in the development of medical treatment protocols that are personalized to individual soldiers' needs, improving recovery times and outcomes in battlefield injuries. Additionally, AI will enhance medical logistics, ensuring that critical supplies and equipment are available when and where they are needed.

7.10.8 Future Directions for the General Application of AI in Defense

Predictive systems in defense will see significant advancements as AI continues to improve the accuracy of forecasting enemy movements, equipment maintenance needs, and logistical requirements. Future innovations will enable predictive systems to process larger datasets and refine their algorithms, resulting in more accurate and timely predictions. These systems will not only predict battlefield outcomes but also assist in planning logistics, optimizing resource allocation, and anticipating humani-

tarian needs in conflict zones, thereby enhancing overall operational effectiveness.

In decision support systems, AI will play a critical role in augmenting human decision-making by providing clear, data-driven insights that reduce uncertainty in complex situations. Future AI-driven decision support systems will leverage advanced NLP and machine reasoning to provide more intuitive and interactive interfaces, enabling decision-makers to engage with AI recommendations more effectively. These systems will also incorporate explainability features, ensuring that human operators understand the rationale behind AI-driven recommendations, which is crucial in high-stakes scenarios such as military engagements or crisis response.

7.11 Conclusion

AI has proven to be highly effective across a wide range of defense applications, fundamentally transforming how military operations and training are conducted. In C2 systems, AI enhances situational awareness and decision-making by processing vast amounts of data in real time, allowing commanders to respond more quickly and accurately to changing conditions. Autonomous systems, such as uncrewed aerial and ground vehicles, have significantly reduced the need for human involvement in dangerous missions, minimizing risks to personnel while improving operational efficiency. In cybersecurity, AI has become an indispensable tool, enabling the detection and neutralization of sophisticated cyber threats at speeds far beyond human capability. Additionally, AI-driven training and simulation systems provide more personalized and immersive training experiences, improving readiness and the transfer of skills to real-world operations.

Despite its clear advantages, the effectiveness of AI in defense also hinges on addressing challenges such as bias, accountability, and transparency. We need AI solutions that we can trust. While AI systems offer increased speed and accuracy in decision-making, human oversight remains critical to ensuring that these technologies operate within ethical and legal frameworks, particularly in areas such as autonomous weapons and cyber warfare. The integration of XAI and rigorous testing of AI models can

help mitigate these risks, ensuring that AI remains a reliable tool in military operations. As AI continues to evolve, its role in defense will only expand, offering greater strategic advantages while pushing the boundaries of innovation and operational capability. Achieving the full potential of AI will require ongoing collaboration between governments, industry, and academia to ensure responsible deployment and continual improvement in AI systems.

References

Aguirre, B.V., Uresti, J.A.R., and Boulay, B.D. (2016). An analysis of student model portability. *International Journal of Artificial Intelligence in Education* 26 (3): 932–974. https://doi.org/10.1007/s40593-016-0113-0.

Akpan, B. (2020). Classical and operant conditioning—Ivan Pavlov; Burrhus Skinner. In: *Science Education in Theory and Practice: An Introductory Guide to Learning Theory* (ed. B. Akpan and T. J. Kennedy), pp. 71–84. Springer International Publishing. https://doi.org/10.1007/978-3-030-43620-9_6.

Alberts, D.S. and Hayes, R.E. (2006). *Understanding Command and Control*. CCRP Publication Series.

Alfehaid, A. and Hammami, M. (2023). Artificial intelligence in education: literature review on the role of conversational agents in improving learning experience. *International Journal* 10 (3): 3121–3129.

Alsawaier, R.S. (2018). The effect of gamification on motivation and engagement. *The International Journal of Information and Learning Technology* 35 (1): 56–79. https://doi.org/10.1108/IJILT-02-2017-0009.

Arrieta, A.B., Díaz-Rodríguez, N., Del Ser, J., et al. (2020). Explainable artificial intelligence (XAI): concepts, taxonomies, opportunities and challenges toward responsible AI. *Information Fusion* 58: 82–115.

Barber, D., Reinerman-Jones, L., Wohleber, R., et al. (2022). Multimodal, adaptable, and dynamic human autonomy team relationships. In: *Interservice/Industry Training, Simulation and Education Conference (I/ITSEC)*. Orlando, FL.

Bratman, M.E. (1987). *Intentions, Plans, and Practical Reason*. Harvard University Press.

Bunch, K., Hou, A.C., Haberman, R., et al. (2024). *Risk Assessment of Reinforcement Learning AI Systems: Looking Beyond the Technology*.

RAND Corporation.

Choi, G.W., Kim, S.H., Lee, D., et al. (2024). Utilizing generative AI for instructional design: exploring strengths, weaknesses, opportunities, and threats. *TechTrends* 68: 1–13.

Colby, B.R. (2020). From systems to services: changing the way we conceptualize ITSs—a theoretical framework and proof-of-concept. Doctoral dissertation. Brigham Young University.

Colby, B.R., Tucker, E., and Siggins, T. (2024). Beyond standalone systems: creating an ecosystem of adaptive training services. In: *International Conference on Human–Computer Interaction* (pp. 3–14). Switzerland: Springer Nature.

DARPA (2017). *DARPA perspective on AI (archived)*. https://www.darpa.mil/about-us/darpa-perspective-on-ai.

DARPA (2024). *Enhancing design for graceful extensibility*. https://www.-darpa.mil/program/enhancing-design-for-graceful-extensibility.

Department of Defense (2020). *Joint publication 5-0 joint operation planning*. https://irp.fas.org/doddir/dod/jp5_0.pdf.

Doyle, J. (1979). A truth maintenance system. *Artificial Intelligence* 12 (3): 231–272.

Elshtain, J.B. (ed.) (1992). *Just War Theory*. NYU Press.

Engelbart, D.C. (2023). Augmenting human intellect: a conceptual framework. In: *Augmented Education in the Global Age* (pp. 13–29). Routledge.

Federal Aviation Administration (FAA) (2024). Updates on Boeing 737 MAX. https://www.faa.gov/newsroom/faa-updates-boeing-737-max-0.

Folsom-Kovarik, J.T., Roque, A., and Sinatra, A.M. (2022). Addressing team process with automated speech act assessments. In *Proceedings of the Tenth Annual GIFT User Symposium* (pp. 139–146).

Franklin, S. and Graesser, A. (1996). Is it an agent, or just a program? A taxonomy for autonomous agents. In *International Workshop on Agent Theories, Architectures, and Languages* (pp. 21–35). Berlin Heidelberg: Springer.

García-Salinas, J.S., Villaseñor-Pineda, L., Reyes-García, C.A., et al. (2019). Transfer learning in imagined speech EEG-based BCIs. *Biomedical Signal Processing and Control* 50: 151–157.

Georgeff, M., Pell, B., Pollack, M., et al. (1999). The belief-desire-intention model of agency. In *Intelligent Agents V: Agents theories, Architectures,*

and Languages: 5th International Workshop, ATAL'98 Paris, France, July 4–7, *1998 Proceedings 5* (pp. 1–10). Berlin Heidelberg: Springer.

Green, A.C., Johnson, M.T., and Davis, E.W. (2020). Leveraging adaptive learning technologies for personalized training: a review of real-time data collection techniques. *Educational Technology Research and Development* 68 (2): 331–349. https://link.springer.com/journal/11423.

Gruber, M.E., Hancock, P.A., Barber, D.J., et al. (2024). The impact of transparency on human-autonomy teaming. In *Proceedings of the Human Factors and Ergonomics Society Annual Meeting* (p. 10711813241277519). SAGE Publications.

Hayes-Roth, B. (1995). An architecture for adaptive intelligent systems. *Artificial Intelligence* 72 (1–2): 329–365.

Heinze, C., Goss, S., Josefsson, T., et al. (2002). Interchanging agents and humans in military simulation. *AI Magazine* 23 (2): 37–37.

Hung, V. et al. (2020). Towards hypoxia detection via automatic speech recognition technology: a theoretical technology framework. In *SAFE Association 62nd Annual Symposium* (Virginia Beach, VA, 2020).

Hurd, W.L., Harris, P.D.G., and Smith, S.L. (2019). Iterative refinement in training programs: the role of feedback and evaluation. *Journal of Training and Development* 15 (3): 221–238. https://onlinelibrary.wiley.-com/journal/14682419.

Jensen, B. and Tadross, D. (2023). How large-language models can revolutioninze military planning. *War on the rocks*. https://warontherocks.-com/2023/04/how-large-language-models-can-revolutionize-military-planning/ (accessed 7 September 2024).

Johnson, S.E. and Levis, A.H. (1989). *Science of Command and Control: Part II, Coping with Complexity*. AFCEA International Press.

Jones, R.M. and Wray, R.E. (2006). Comparative analysis of frameworks for knowledge-intensive agents. *AI Magazine* 27 (2): 45–56.

Kakar, S., Maiti, P., Taneja, K., et al. (2024, June). Jill Watson: scaling and deploying an AI conversational agent in online classrooms. In *International Conference on Intelligent Tutoring Systems* (pp. 78–90). Cham, Switzerland: Springer Nature.

Kawatsu, C., Jones, R., McLellan, C. (2020). *Final Report for Alpha Dogfight Trials*. Air Combat Evolution (ACE) Program. DARPA Strategic Technology Office.

Kawatsu, C., Koss, F., Gillies, A., et al. (2017a, August). Gesture recognition for robotic control using deep learning. In *NDIA Ground Vehicle Systems Engineering and Technology Symposium*. Novi, Michigan.

Kawatsu, C., Sperlich, C., Stensrud, B. (2017b). *DeepAgent: Phase I SBIR Base Period Final Report*. U.S. Navy Contract N68335-17-C-0179.

Kotseruba, I. and Tsotsos, J.K. (2020). 40 years of cognitive architectures: core cognitive abilities and practical applications. *Artificial Intelligence Review* 53 (1): 17–94.

Kumar, S.P., Lee, R.J., and Patel, T.K. (2022). Machine learning approaches for performance benchmarking in training programs. *Machine Learning in Education* 14 (1): 45–60. https://link.springer.com/journal/40940.

Laird, J.E. (2012). *The Soar Cognitive Architecture*. MIT Press.

Lashbrook, A. (2020). *Duolingo needs to chill*. Debugger. https://debugger.medium.com/duolingo-needs-to-chill-8f1832745ca0.

Lasko, J., Ma, J., Nicoletti, M., et al. (2024). Using i-vectors for subject-independent cross-session EEG transfer learning. *arXiv preprint*: https://arxiv.org/abs/2401.08851.

Lebanoff, L., Paul, N., Ballinger, C., et al. (2023). A comparison of behavior cloning methods in developing interactive opposing-force agents. In *The International Flairs Conference Proceedings* (Vol. 36, no. 1). https://doi.org/10.32473/flairs.36.133299.

Lee, J. and Park, O.C. (2008). Adaptive instructional systems. In *Handbook of Research on Educational Communications and Technology* (pp. 469–484). Routledge.

Leiker, D., Finnigan, S., Gyllen, A.R., et al. (2023). Prototyping the use of large language models (LLMs) for adult learning content creation at scale. *arXiv preprint*: https://arxiv.org/abs/2306.01815.

Madunić, J. and Sovulj, M. (2024). Application of ChatGPT in Information literacy instructional design. *Publications* 12 (2): 11.

Mcsorley, J. (2022). Effects of gamification on knowledge acquisition: aviation weather online training. Doctoral Dissertations and Master's Theses. 646. https://commons.erau.edu/edt/646.

Montasari, R. (2023). National artificial intelligence strategies: a comparison of the UK, EU and US approaches with those adopted by state adversaries. In *Countering Cyberterrorism: The Confluence of Artificial*

Intelligence, Cyber Forensics and Digital Policing in US and UK National Cybersecurity (pp. 139–164). Springer International Publishing.

Mulian, H., Shlomov, S., Limonad, L., et al. (2024). Mimicking the maestro: exploring the efficacy of a virtual AI teacher in fine motor skill acquisition. In *Proceedings of the AAAI Conference on Artificial Intelligence* (Vol. 38. no. 21).

National Geospatial-Intelligence Agency (NGA) (2024). *NGA publishes draft GEOINT research methods topic to the boosting innovative GEOINT research BAA*. https://www.nga.mil/news/NGA_publishes_Draft_GEOINT_research_methods_Topic_.html.

Newell, A. (1992). Précis of unified theories of cognition. *Behavioral and Brain Sciences* 15 (3): 425–437.

Newton, C., Ballinger, C., Sloma, M., et al. (2022). Hierarchical, discontinuous agent reinforcement learning rewards in complex military-oriented environments. In *The International FLAIRS Conference Proceedings* (Vol. 35).

Noh, D. (2020). The gamification framework of military flight simulator for effective learning and training environment. University of Central Florida.

O'Donoughue, N.A., McBirney, S., and Persons, B. (2021). *Distributed kill chains: drawing insights for mosaic warfare from the immune system and from the navy*. RAND Corporation.

Prasongpongchai, T., Pataranutaporn, P., Kanapornchai, C., et al. (2024). Interactive AI-generated virtual instructors enhance learning motivation and engagement in financial education. In *International Conference on Artificial Intelligence in Education*. Switzerland: Springer Nature.

Program Executive Office, Defense Healthcare Management Systems (2025). Military Health System. https://www.health.mil/Reference-Center/Fact-Sheets/2025/04/23/Program-Executive-Office-Defense-Healthcare-Management-Systems-Fact-Sheet.

Rehm, J., Reshodko, I., Børresen, S.Z., et al. (2024). The virtual driving instructor: multi-agent system collaborating via knowledge graph for scalable driver education. In *Proceedings of the AAAI Conference on Artificial Intelligence* (Vol. 38, No. 21, pp. 22806–22814).

Rudin, C. (2019). Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence* 1 (5): 206–215.

Sauter, J. and Bixler, K. (2021). Design of unmanned swarm tactics for an urban mission. In *Proceedings Volume 11021 Unmanned Systems Technology XXI* (ed. C.M. Shoemaker, H.G. Nguyen, and P.L. Muench). SPIE Defense + Commercial Sensing 2019.

Sheth, A. and Thirunarayan, K. (2021). The duality of data and knowledge across the three waves of AI. *IT Professional* 23 (3): 35–45.

Sottilare, R., McGroarty, C., Ballinger, C., et al. (2023). Investigating the effect of realistic agents on team learning in adaptive simulation-based training environments using GIFT. In *Proceedings of the 11th Annual Generalized Intelligent Framework for Tutoring (GIFT) Users Symposium (GIFTSym11)*. US Army Combat Capabilities Development Command–Soldier Center.

Sottilare, R. and DeFalco, J. (2020). *Fundamentals of adaptive instruction systems tutorial*. I/ITSEC.

Spain, R., Rowe, J., Goldberg, B., et al. (2019). *Enhancing learning outcomes through adaptive remediation with GIFT*. I/ITSEC.

Sussman, E.S., Chen, S., Sussman-Fort, J., et al. (2014). The five myths of MMN: redefining how to use MMN in basic and clinical research. *Brain Topography* 27: 553–564.

Sussman-Fort, J., Glenister, L., Jankowski, D., et al. (2025). Prototyping intelligent software solutions for global health engagement. Military Medicine, usaf164.

Tanaka, A., Craighead, J., Taylor, G., et al. (2019). Adaptive learning technology for AR training: possibilities and challenges. In *Adaptive Instructional Systems: First International Conference, AIS 2019, Held as Part of the 21st HCI International Conference, HCII 2019*, Orlando, FL, USA, July 26–31, *2019, Proceedings 21* (pp. 142–150). Springer International Publishing.

Tanaka, A., Sussman-Fort, J., and Cohn, J. (2024). Automated decision support for remote health monitoring—special forces autonomous vital injury observation and rescue (SAVIOR). 2024 Air Force Research Laboratory (AFRL) Biotech Days Summit. Wright State University, Dayton, Ohio. https://www.ues.com/hubfs/Biotech%20Days%20Program-1.pdf.

Tangredi, S.J. and Galdorisi, G.V. (2021). *AI at War: How Big Data Artificial Intelligence and Machine Learning are Changing Naval Warfare*. Naval Institute Press.

Taylor, G., Craighead, J., Menefee, K., et al. (2024). An interactive virtual assistant for flexible just-in-time training. *Human Factors in Design, Engineering, and Computing* 159 (159)

Taylor, G. and Sims, E. (2009). Developing believable interactive cultural characters for cross-cultural training. In *Online Communities and Social Computing* (ed. A.A. Ozok and P. Zaphiris), pp. 282–291. Springer. https://doi.org/10.1007/978-3-642-02774-1_31.

Ulinski, M., Medjuck, E., and Phillips, H. (2024). Automated radio operator utterance recognition for US Navy training. In *Proceedings of Interservice/Industry Training, Simulation, and Education Conference (I/ITSEC)*. Orlando, FL (in press).

Watanabe, H., Tanaka, H., Sakti, S., et al. (2020). Synchronization between overt speech envelope and EEG oscillations during imagined speech. *Neuroscience Research* 153: 48–55.

Wilson, R.A., Thompson, J.E., and Morgan, L.H. (2021). Benchmarking training programs with artificial intelligence: methodologies and applications. *International Journal of Artificial Intelligence in Education* 31 (4): 456–472. https://link.springer.com/journal/40593.

Woods, D.D. (2018). The theory of graceful extensibility: basic rules that govern adaptive systems. *Environment Systems and Decisions* 38 (4): 433–457.

Wray, R.E., Folsom-Kovarik, J.T., Woods, A., et al. (2015). Motivating narrative representation for training cross-cultural interaction. *Procedia Manufacturing* 3: 4121–4128. https://doi.org/10.1016/j.promfg.2015.07.985.

Zhang, T., Qin, Y., and Li, Q. (2021). Trusted artificial intelligence: technique requirements and best practices. In *2021 International Conference on Cyberworlds (CW)* (pp. 303–306). IEEE.

Zuluaga-Gomez, J., Prasad, A., Nigmatulina, I., et al. (2023). A virtual simulation-pilot agent for training of air traffic controllers. *Aerospace* 10 (5): 490.